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DEPARTMENT OF AGRICULTURE

SOIL SCIENCE

COURSE CODE: **NR 240 TEST 1**

INSTRUCTIONS: ANSWER ALL QUESTIONS.

TIME 1.5 HRS

Question 1

Define the following;

(i) Atmospheric Electric Phenomenon ***(2 marks)***

(ii) Oxidation ***(2 marks)***

(iii) Metamorphic Rocks ***(2 marks)***

(iv) Pedologic Time ***(2 marks)***

(v) Polypedon ***(2 marks)***

Question 2

Explain the role of living organisms and climate in soil formation. ***(10 marks)***

Question 3

Most of Zambia's plateau and some hilly areas have been degraded due to geomorphological processes. Zambia has three main agro-ecological zones that comprise of different soil types. In brief outline the short falls and strength's associated with the soils of the three regions. ***(10 marks).***

Question 4

Briefly answer the following questions as listed below?

(i) What is the composition of atmospheric nitrogen by volume (%) and weight (%) basis? ***(4 marks)***

(ii) Define and give **two** examples for each of the following, the extrusive and intrusive Igneous Rocks ***(4 marks)***

(iii) List any **six** properties of Soil Minerals. ***(3 marks)***

(iv) What are the major characteristics of the **A** and **E** horizons of the soil profile? ***(3 marks)***

(v) Compare and contrast the chemical weathering processes of ***Hydration*** and ***Solution***. And illustrate your answer with the appropriate chemical equations. ***(6 marks)***

END

KAPASA MAKASA UNIVERSITY

DEPARTMENT OF AGRICULTURE AND AQUATIC SCIENCES

SOIL SCIENCE

COURSE CODE: **NR 240 TEST II-MAKE UP**

INSTRUCTIONS: ANSWER ALL QUESTIONS.

TIME 1.5 HRS

Question 1

Define the following;

(i) Porosity *(2 marks)*
(ii) Hue *(2 marks)*
(iii) Silt *(2 marks)*
(iv) Grade of Soil Structure *(2 marks)*
(v) Aggregate Instability Index (AII) *(2 marks)*

Question 2

What is the Interpretation of Soil Bulk Density values in Soil Health? *(10 marks)*

Question 3

Calculate the porosity of a 250 g sample that contains 65 g of water when 55% of the pores are full of water. *(10 marks).*

Question 4

Calculate the bulk density of a soil sample that has a porosity of 45.5%. (Assume Particle density of 2.65 g/cm^3). *(10 marks)*

Question 5

Briefly answer the following questions as listed below?

(i) What is the significance of knowing the structure of the soil for Agroforestry scholars? *(4 marks)*
(ii) Define and give two ways on how the water content in the soil can be expressed? *(6 marks)*

END

KAPASA MAKASA UNIVERSITY

DEPARTMENT OF AGRICULTURE AND AQUATIC SCIENCES

SOIL SCIENCE

COURSE CODE: NR 240 TEST 3 TIME: 1.5 HRS

INSTRUCTIONS: ANSWER ALL QUESTIONS.

Question 1

Define the following;

(i) Potential Hydrogen (pH) ***(2 marks)***

(ii) Soil Colloid ***(2 marks)***

(iii) Isormophous Substitution ***(2 marks)***

(iv) Cation Exchange Capacity (CEC) ***(2 marks)***

(v) Chemoautotrophs ***(2 marks)***

Question 2

Discuss the functions of soil micro-organisms in relation to soil health and fertility ***(10 marks)***

Question 3

(a) How do parent material, vegetation and climate affect the pH soil? ***(6 marks)***

(b) A soil has a pH (0.01 M CaCl2) of 4.22, what is the agronomic interpretation of the pH of this soil? ***(4 marks)***

Question 4

A soil from Kapasa Makasa University (KMU) contains 36 % clay and has a CEC of 1.7 cmol (+)/kg soil. Calculate the CEC of the clay fraction assuming the measured CEC is due to clay only and ***state*** the likely type of a soil colloid. ***(10 marks)***

Question 5

(a) Microorganisms are classified into two broad classes with respect to their carbon and energy sources. The two broad classes are heterotrophs and autotrophs.Dintinguish between the ***two*** classes. **(5 marks)**

(b) The soil contains ***five major*** groups of microorganisms that help in soil health and fertility. List them? **(5 marks)**

END